

2023 HKDSE Physics

Paper 1A — Multiple-choice Questions

Original questions and official answers with detailed English reasoning. Calculations show the governing relationships, substitution and units; each structured subpart is presented separately.

Editable questions with original diagrams, official answers and detailed English reasoning. Each subpart is separated for clarity.

Questions

Official answers are placed directly below each question.

Question 1

Which of the following statements involving heat transfer via radiation is/are correct?

- (1) Objects hotter than the surroundings do not absorb heat by radiation.
- (2) The silvery surfaces of survival blankets help the body absorb heat from the surroundings by radiation.
- (3) The cooling fins of a car engine should preferably be black in colour.

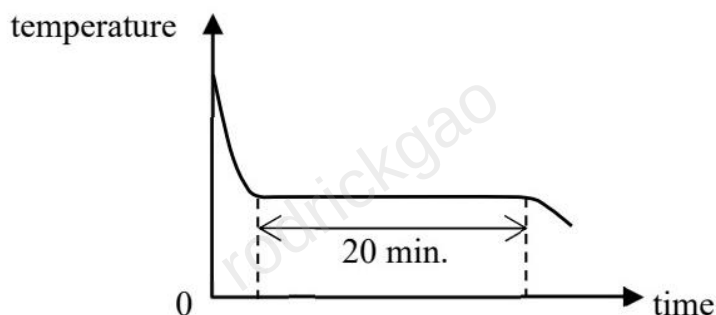
- A. (1) only
- B. (3) only**
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: B

Every object both emits and absorbs thermal radiation; a hotter object has a net outward transfer, not zero absorption. A silvery survival blanket reduces radiative heat loss from the body. Dull black cooling fins are good emitters, so only (3) is correct. Answer: B.

Question 2

A hot liquid substance of mass 0.3 kg and with heat capacity $600 \text{ J } ^\circ\text{C}^{-1}$ is left to cool in air. Its temperature falls at a rate of 2°C per minute just before it begins to solidify. Then its temperature remains steady for 20 minutes until all the liquid just solidifies. Estimate the specific latent heat of fusion of the substance.



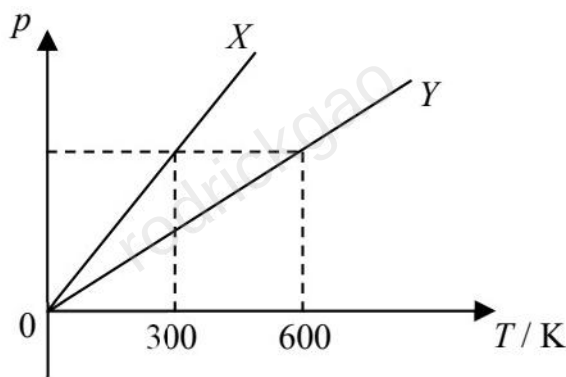
- A. 20000 J kg^{-1}
- B. 24000 J kg^{-1}
- C. 48000 J kg^{-1}
- D. 80000 J kg^{-1}

Official answer: D

The heat-loss rate just before freezing is $P = C|\Delta T/\Delta t| = 600(2/60) = 20 \text{ W}$. Estimate the same rate during the constant-temperature plateau. The latent heat released is $Q = Pt = 20(20 \times 60) = 24000 \text{ J}$, so $l = Q/m = 24000/0.3 = 80000 \text{ J kg}^{-1}$. Answer: D.

Question 3

A fixed mass of an ideal gas contained in a closed vessel X of volume V is heated. Line X in the figure shows the variation of pressure p with temperature T of the gas. The experiment is repeated with another closed vessel Y containing the same amount of this ideal gas and line Y represents the result. Find the volume of vessel Y.



- A. 0.25 V
- B. 0.5 V
- C. 2 V
- D. 4 V

Official answer: C

For the same amount of ideal gas, $pV = nRT$. At the pressure marked by the dashed horizontal line, $T_X = 300$ K and $T_Y = 600$ K. Hence $V_Y/V_X = T_Y/T_X = 2$, so $V_Y = 2V$. Equivalently, the p - T slope is nR/V and Y has half the slope. Answer: C.

Question 4

Which of the following are vector quantities?

- (1) acceleration
- (2) momentum
- (3) work done

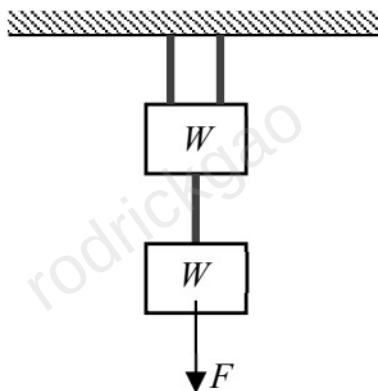
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: A

Acceleration and momentum have both magnitude and direction and combine by vector addition. Work is a scalar: $W = Fs \cos\theta$, so a positive or negative value does not make it a vector. Thus (1) and (2) are correct. Answer: A.

Question 5

Two blocks, each of weight W , are hung by three identical light strings as shown. Each string has a limiting tension of $2W$, i.e. it will break for a tension larger than this value. Force F acts vertically on the lower block. What is the largest value of F without breaking any of the strings?



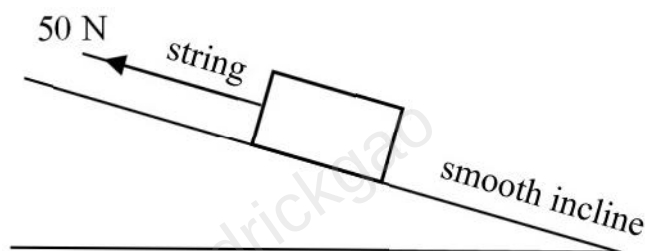
- A. $0.5W$
- B. W
- C. $1.5W$
- D. $3W$

Official answer: B

The lower string supports W plus the downward applied force, so its tension is $W+F$. The two identical upper strings share $2W+F$ equally, giving $W+F/2$ each. The lower string reaches its $2W$ limit first: $W+F = 2W$, so $F_{\max} = W$. Each upper tension is then $1.5W$, still below its limit. Answer: B.

Question 6

On a smooth incline, a block of weight 120 N is kept stationary by a string with tension 50 N parallel to the incline. The string is then cut and the block accelerates down the incline. What is the net force acting on the block before and after the string is cut?



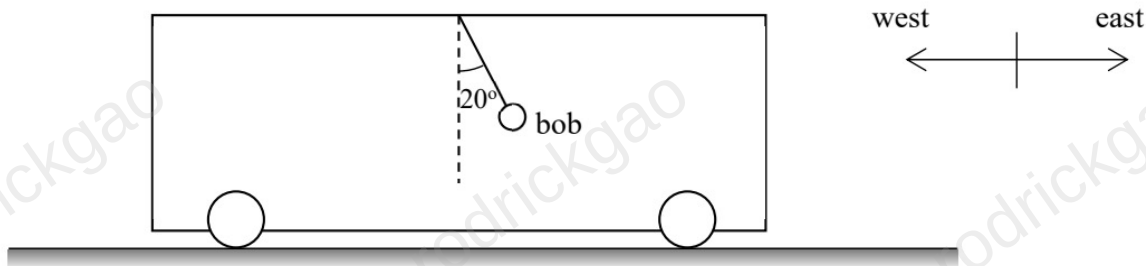
- A. Before: 50 N; after: 70 N
- B. Before: 50 N; after: 50 N
- C. Before: 0 N; after: 70 N
- D. Before: 0 N; after: 50 N

Official answer: D

Before cutting, the block is stationary, so its net force is zero. The 50 N tension balances the component of weight down the plane. Cutting removes the tension but does not change that component; the normal reaction still balances the perpendicular component. The new net force is 50 N down the plane. Answer: D.

Question 7

A bob is suspended from the ceiling of a train moving along a horizontal east-west rail. The bob slants towards east and makes an angle of 20° with the vertical as shown. What is the direction and magnitude of the acceleration of the train? ($g = 9.81 \text{ m s}^{-2}$)



- A. Due west; 3.36 m s^{-2}
- B. Due west; 3.57 m s^{-2}**
- C. Due east; 3.36 m s^{-2}
- D. Due east; 3.57 m s^{-2}

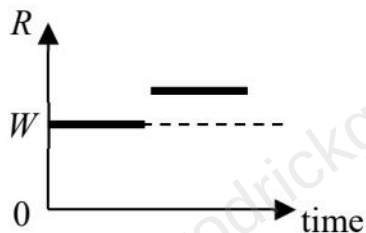
Official answer: B

The string tension points up and west. Vertically, $T \cos 20^\circ = mg$; horizontally, $T \sin 20^\circ = ma$. Therefore $a = g \tan 20^\circ = 9.81 \tan 20^\circ = 3.57 \text{ m s}^{-2}$ due west. The eastward tilt indicates acceleration westward, not necessarily motion westward. Answer: B.

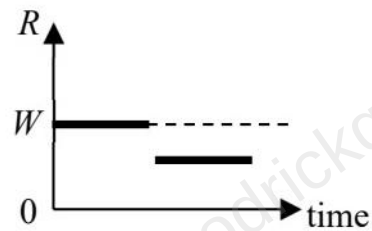
Question 8

In an amusement park ride, a boy of weight W undergoes free fall vertically in a chair at first and then decelerates uniformly. Which graph below correctly shows the time variation of the reaction force R acting on the boy by the chair? Neglect air resistance.

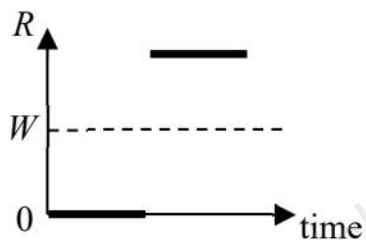
A.



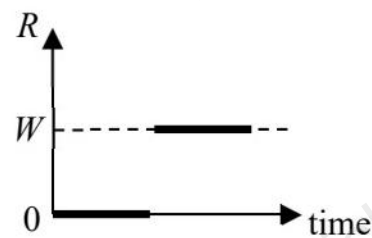
B.



C.



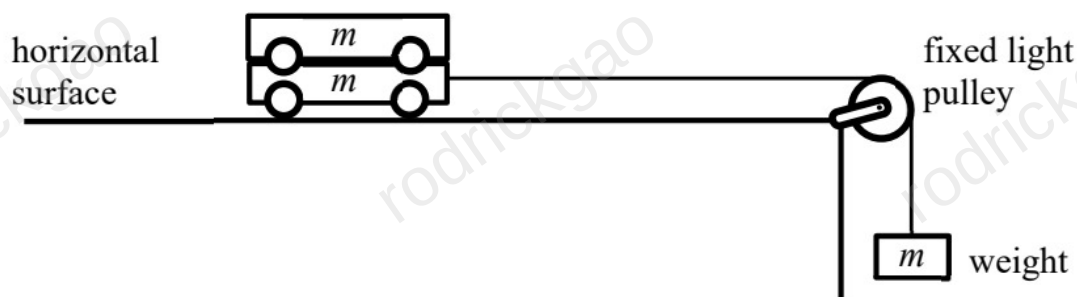
D.

**Official answer: C**

During free fall the boy and chair accelerate downwards at g , so $R = 0$. During uniform slowing of the downward motion, acceleration is constant upwards. Thus $R - W = ma$ and $R = W + ma$, a constant greater than W . Graph C shows these two stages. Answer: C.

Question 9

The figure shows a set-up in which a weight of mass m is attached to two trolleys stacked together, each trolley of mass m , via a light inextensible string. After the weight is released from rest, which of the following is correct? Assume that all contact surfaces are smooth and neglect air resistance. g is the acceleration due to gravity.



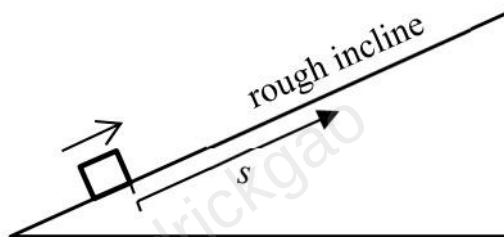
- A. Acceleration: $g/2$; tension: smaller than mg
- B. Acceleration: $g/2$; tension: equal to mg
- C. Acceleration: $g/3$; tension: smaller than mg
- D. Acceleration: $g/3$; tension: equal to mg

Official answer: C

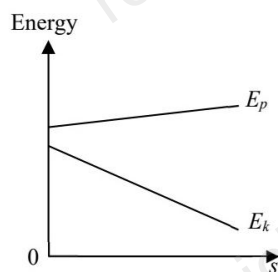
The two trolleys shown are both attached to the string, so the accelerating mass is $3m$. Along the motion, the hanging weight supplies the driving force mg : $mg = 3ma$, giving $a = g/3$. For the hanging mass, $mg - T = ma$, hence $T = 2mg/3 < mg$. Answer: C.

Question 10

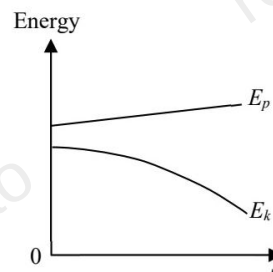
A block is projected upwards along a rough incline which has a constant frictional force acting on the block. Which of the following correctly shows the variation of potential energy E_p and kinetic energy E_k with distance s that the block travelled up the incline? Neglect air resistance.



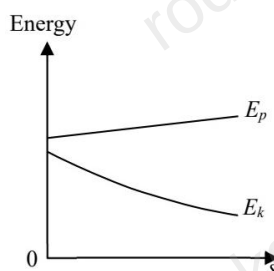
A.



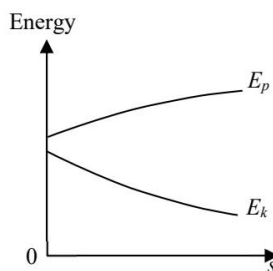
B.



C.



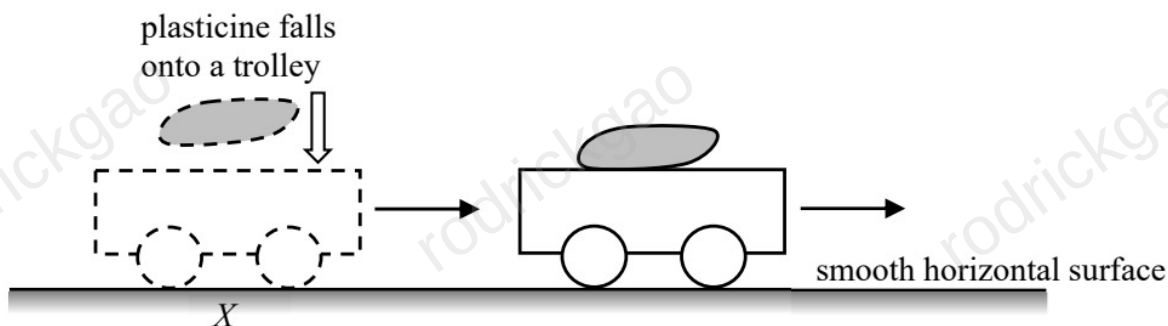
D.

**Official answer: A**

On a straight incline, height increases linearly with distance s , so $E_p = E_{p0} + mg s \sin\theta$. With constant friction f , the work–energy theorem gives $E_k = E_{k0} - (mg \sin\theta + f)s$. Both graphs are straight, and the kinetic-energy decrease is greater than the potential-energy increase. This is graph A. Answer: A.

Question 11

A trolley moves with a constant velocity along a smooth horizontal surface. When the trolley reaches point X, a plasticine falls vertically onto it. After the collision, they stick together and continue to move forward. Which description about the total linear momentum of the trolley and the plasticine just before and just after collision is correct?



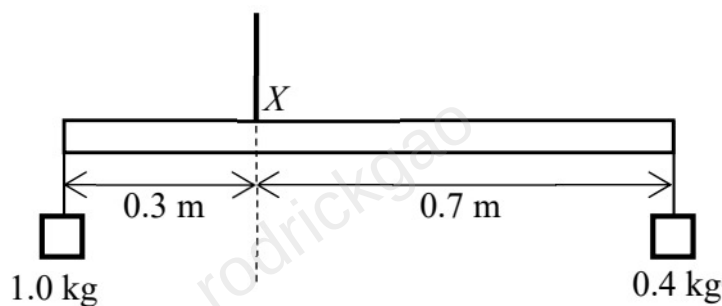
- A. Horizontal: conserved; vertical: conserved
- B. Horizontal: conserved; vertical: not conserved
- C. Horizontal: not conserved; vertical: conserved
- D. Horizontal: not conserved; vertical: not conserved

Official answer: B

There is no external horizontal impulse on the trolley–plasticine system, so horizontal momentum is conserved. During impact the ground supplies an external vertical impulse: the plasticine loses its downward momentum and the combined system has no vertical motion afterwards. Vertical momentum of this system is therefore not conserved. Answer: B.

Question 12

A uniform metre rule of a certain weight is hung by a string at X such that it is kept in equilibrium with weights hanging at both ends as shown. If the 1.0-kg weight is shifted 0.1 m towards X, what distance should the 0.4-kg weight be shifted towards X in order to restore equilibrium?



- A. 0.10 m
- B. 0.20 m
- C. 0.25 m
- D. 0.45 m

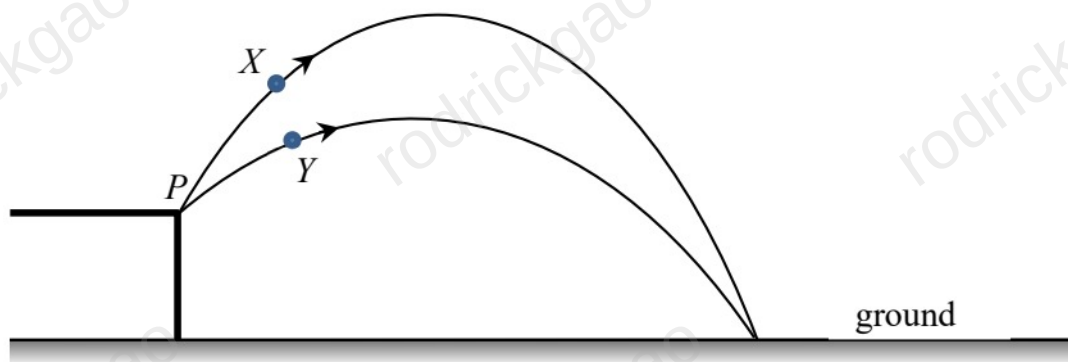
Official answer: C

The rule's weight and its moment about X remain unchanged. Moving the 1.0 kg mass reduces its moment by $1.0g(0.10)$. To restore balance, the reduction on the other side must be equal: $0.4g d = 1.0g(0.10)$, so $d = 0.25$ m.
Answer: C.

Question 13

Two identical particles X and Y are projected from point P simultaneously with the same initial speed but with different angles as shown. They finally hit the ground at the same point via different paths. Which of the following statements is/are correct? Neglect air resistance.

- (1) They hit the ground at the same time.
- (2) They hit the ground with the same speed.
- (3) The kinetic energy of Y is greater than that of X at their respective maximum heights.



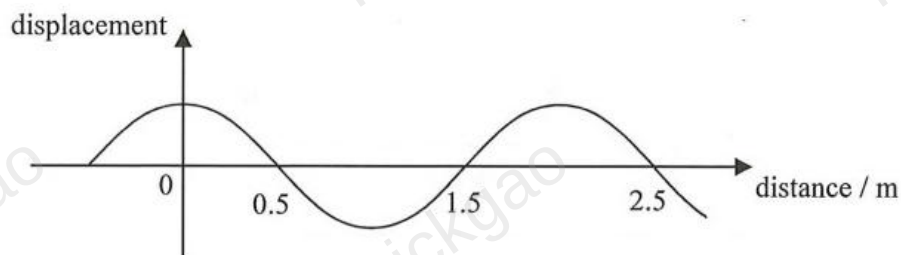
- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

Official answer: D

X has a larger upward component, so it stays in flight longer; (1) is false. Both begin with the same speed and fall through the same height, so conservation of mechanical energy gives equal landing speeds. At the top, only the horizontal velocity remains, and Y's smaller launch angle gives a larger horizontal component and kinetic energy. Thus (2) and (3) are correct. Answer: D.

Question 14

The above figure shows the displacement-distance graph of a wave travelling at a speed of 10 m s^{-1} . What is the period of this wave?



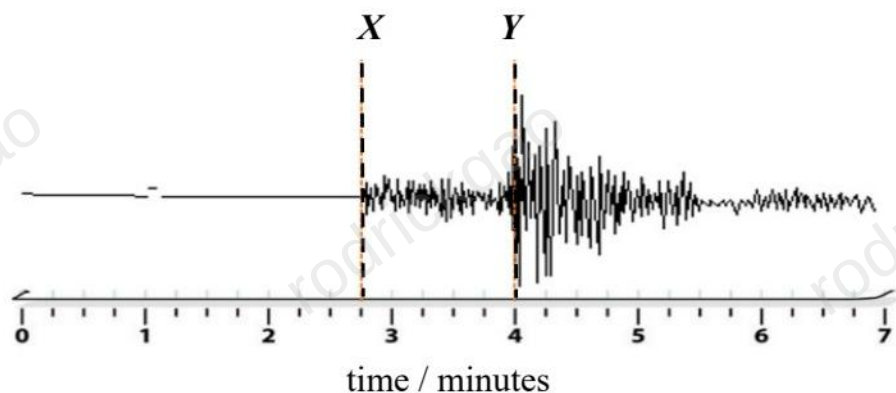
- A. 0.10 s
- B. 0.15 s
- C. 0.20 s
- D. 0.25 s

Official answer: C

The zero crossings at 0.5 m and 2.5 m have the same downward slope, so they are one wavelength apart: $\lambda = 2.0$ m. From $v = \lambda/T$, $T = 2.0/10 = 0.20$ s. Adjacent zero crossings with opposite slopes are only half a wavelength apart. Answer: C.

Question 15

Earthquakes produce P-waves (longitudinal) and S-waves (transverse), which travel at 7.0 km s^{-1} and 4.0 km s^{-1} respectively. On a given day, a seismometer detected the P- and S-waves from a shallow earthquake, with its quake centre located at a distance D from the seismometer. Which deduction below is correct?



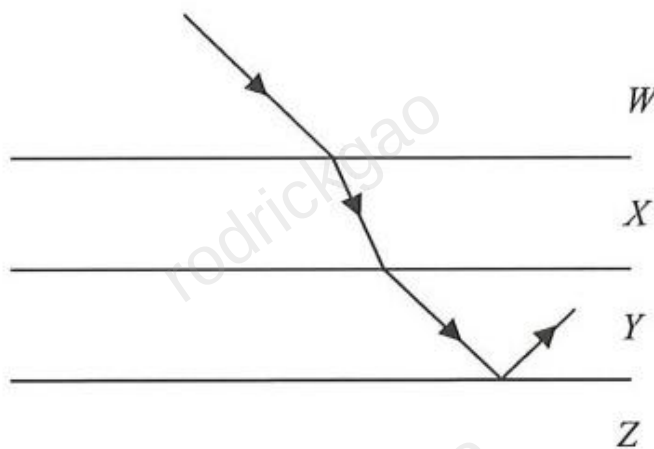
- A. Y: P-waves; $D = 225 \text{ km}$
- B. Y: P-waves; $D = 700 \text{ km}$
- C. Y: S-waves; $D = 225 \text{ km}$
- D. Y: S-waves; $D = 700 \text{ km}$

Official answer: D

P-waves are faster and arrive first at X; Y is the S-wave arrival. The graph gives a delay of $1.25 \text{ min} = 75 \text{ s}$.
 With D in kilometres, $D/4.0 - D/7.0 = 75$, giving $D = 700 \text{ km}$. Use the difference of arrival times, since the earthquake's starting time is not shown. Answer: D.

Question 16

The figure shows a light ray passing through the horizontal boundaries between media W, X, Y and Z. The light ray in W is parallel to that entered into Y from X. Total internal reflection occurs at the boundary between Y and Z. Which medium has the highest refractive index?



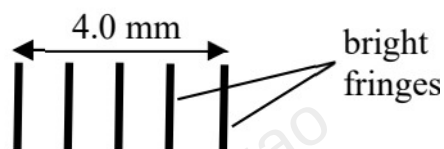
- A. W
- B. X
- C. Y
- D. Z

Official answer: B

At parallel interfaces, $n \sin \theta$ is constant, with θ measured from the normal. The ray has its smallest angle to the normal in X, so X has the largest refractive index. The parallel rays in W and Y give $n_W = n_Y$, while total internal reflection requires $n_Y > n_Z$. Thus $n_X > n_W = n_Y > n_Z$. Answer: B.

Question 17

In Young's double-slit experiment, monochromatic light of wavelength λ is used and the slit separation is a . On a screen placed at a distance of 1.8 m from the double slits, the separation between five consecutive bright fringes is 4.0 mm. Which relationship below is correct?



A. $a = 1800\lambda$

B. $a = 2250\lambda$

C. $a = 7200\lambda$

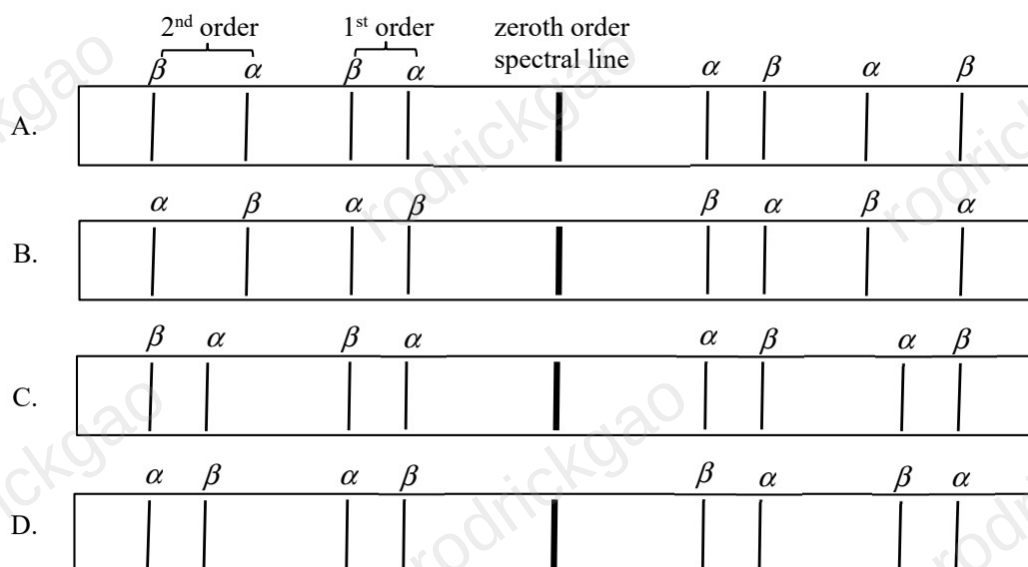
D. $a = 11250\lambda$

Official answer: A

Five consecutive bright fringes span four fringe separations: $\Delta y = 4.0/4 = 1.0$ mm. Young's formula gives $\Delta y = \lambda D/a$, so $a/\lambda = D/\Delta y = 1.8/0.0010 = 1800$. Therefore $a = 1800\lambda$. Answer: A.

Question 18

A diffraction grating is used to observe the α (656 nm) and β (486 nm) lines emitted from a gas discharge tube. Which of the following best represents the pattern observed?

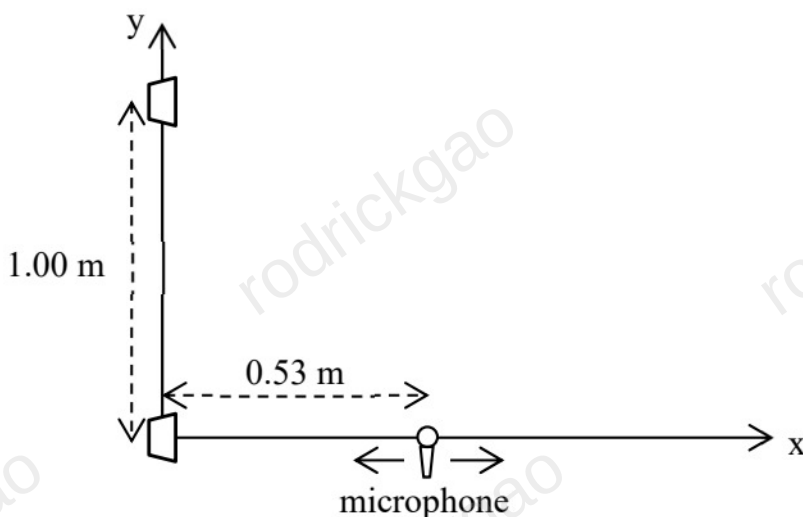


Official answer: B

For a grating, $d \sin \theta = n\lambda$. Within each order, the longer-wavelength α line lies farther from the central maximum than the β line. The separation of the two wavelengths becomes greater in higher orders. Pattern B has both features and is symmetric about the common zeroth-order line. Answer: B.

Question 19

Two loudspeakers separated by 1.00 m along the y-axis emit coherent sound waves which are in phase. When a microphone is moved along the x-axis, only one maximum is detected at a distance of 0.53 m as shown. Find the wavelength of the sound waves.



- A. 0.60 m
- B. 0.75 m
- C. 1.00 m
- D. 1.20 m

Official answer: A

At $x = 0.53$ m, the path difference is $\sqrt{(1.00^2 + 0.53^2)} - 0.53 = 0.6018$ m. It decreases continuously from 1.00 m towards zero as x increases. Since only one finite-distance maximum is found, it is the first-order maximum: $\Delta r = \lambda$. Hence $\lambda \approx 0.60$ m. Answer: A.

Question 20

Virtual images of an object formed by a single lens are always

- (1) erect.
- (2) on the opposite side of the lens to the object.
- (3) smaller than the object.

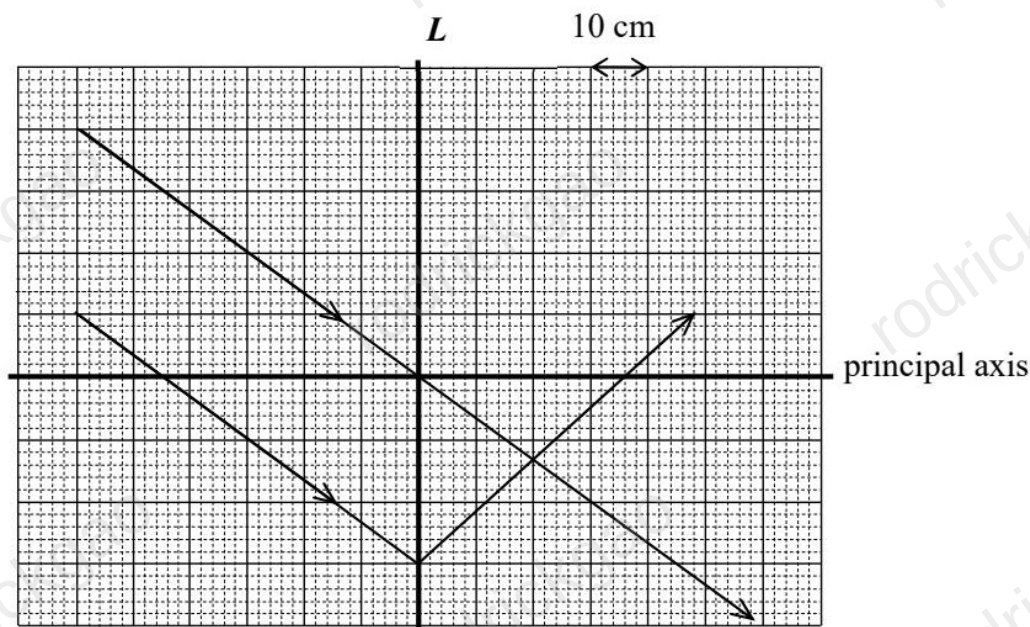
- A. (1) only
- B. (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: A

For a real object, a virtual image produced by a single lens is erect and on the same side as the object. A diverging lens gives a diminished virtual image, while a converging lens with the object inside its focal length gives a magnified virtual image. Only (1) is always true. Answer: A.

Question 21

The ray diagram shows the refraction of two parallel light rays through a lens L. What is the type and the focal length of L?



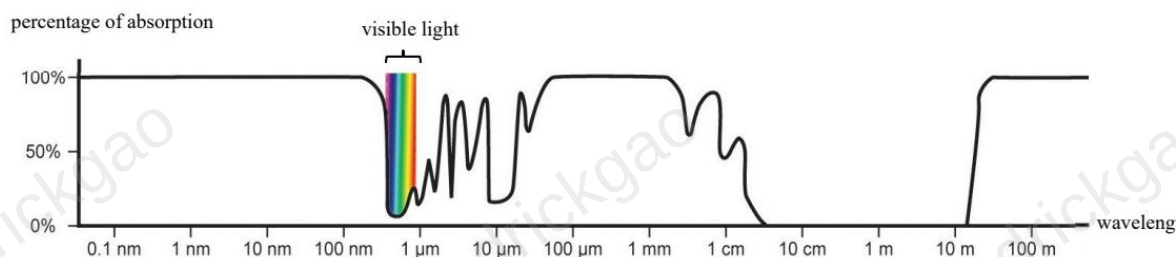
- A. Convex lens; 20 cm
- B. Convex lens; 36 cm
- C. Concave lens; 20 cm
- D. Concave lens; 36 cm

Official answer: A

The refracted rays converge, so L is a convex lens. Obliquely incident parallel rays meet in the focal plane, not necessarily on the principal axis. The horizontal distance from L to their intersection is two of the indicated 10 cm divisions, so $f = 20$ cm. Answer: A.

Question 22

When measuring electromagnetic waves from outer space on the Earth's surface, the percentage of absorption by the atmosphere for various wavelengths is shown below. Referring to this absorption graph, determine which statement is correct.



- A. Gamma rays from galaxies can be observed on Earth's surface.
- B. Nearly all ultra-violet radiation can reach Earth's surface.
- C. Infra-red radiation from stars can reach Earth's surface without absorption.
- D. GHz radio waves (wavelength about 0.5 m) are the best for communications between ground station and space station in the upper atmosphere.

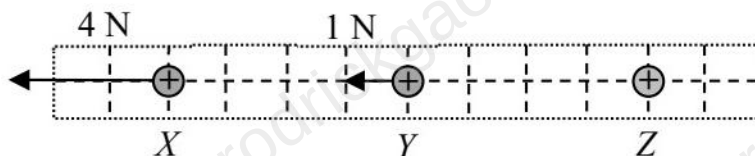
Official answer: D

The graph shows an atmospheric transmission window around $\lambda = 0.5$ m, where absorption is approximately zero. This makes the stated radio waves suitable for ground-space communication. Gamma rays and most ultraviolet are strongly absorbed, and infrared has substantial absorption bands. Answer: D.

Question 23

Positive point charges X, Y and Z are placed along a straight line with Y at the mid-point between X and Z. Assume that the only interaction among them is the electrostatic force. At the instant shown, the net forces acting on X and Y respectively are 4 N and 1 N, both towards the left. Which statements about Z are correct?

- (1) The net force acting on Z is towards the right.
- (2) The net force acting on Z is 5 N in magnitude.
- (3) The charge of Z must be larger than that of X.



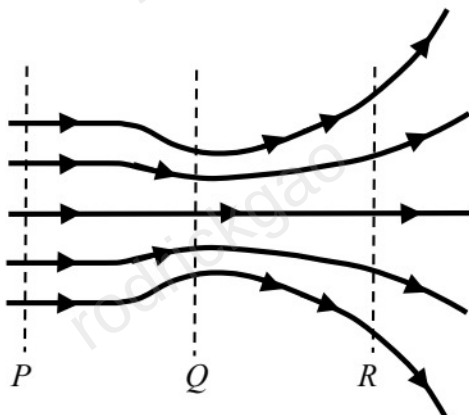
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

Official answer: D

All electrostatic forces are internal to the three-charge system, so their vector sum is zero. With X and Y having net forces of 4 N and 1 N leftwards, Z must have 5 N rightwards. Y is equidistant from X and Z; its net force is leftwards only if Z repels it more strongly, requiring $Q_Z > Q_X$. All three statements are correct. Answer: D.

Question 24

The figure shows the field lines of an electric field. Arrange the respective electric field strengths E_P , E_Q and E_R of regions P, Q and R in ascending order.



A. $E_P < E_Q < E_R$

B. $E_Q < E_R < E_P$

C. $E_R < E_P < E_Q$

D. $E_Q < E_P < E_R$

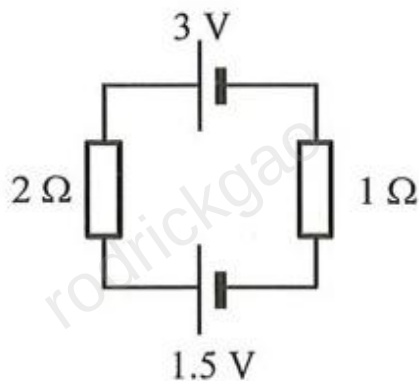
Official answer: C

Electric field strength is greater where field lines are closer together. The lines are most widely spaced at R and most closely spaced at Q, with P intermediate. Hence $E_R < E_P < E_Q$. Answer: C.

Question 25

Two cells of negligible internal resistance are connected to two resistors as shown.

Which of the following is correct?



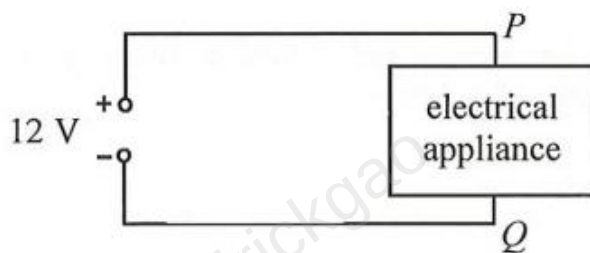
- A. Anticlockwise; 0.5 A
- B. Clockwise; 0.5 A
- C. Anticlockwise; 1.5 A
- D. Clockwise; 1.5 A

Official answer: A

The cells oppose one another, giving a net emf of $3.0 - 1.5 = 1.5$ V. The series resistance is $2 + 1 = 3$ Ω , so $I = 1.5/3 = 0.5$ A. The 3 V cell drives conventional current out of its positive terminal, producing an anticlockwise current. Answer: A.

Question 26

The figure shows an electrical appliance connected to a potential difference of 12 V. When a charge of +2 C passes from P to Q in the circuit, how much electrical energy is supplied to the appliance?



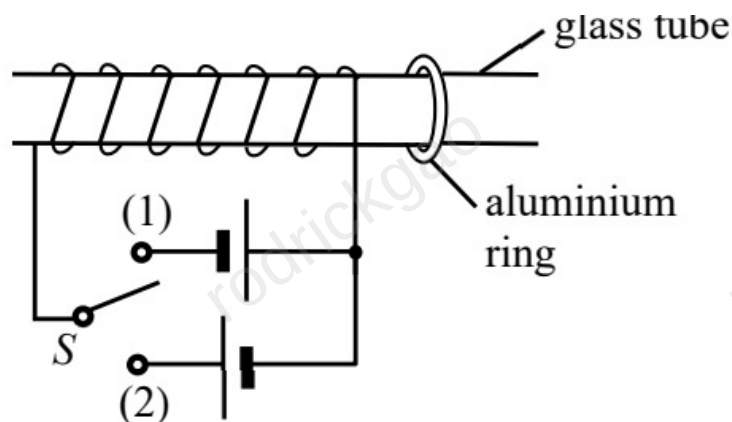
- A. 2 J
- B. 6 J
- C. 12 J
- D. 24 J

Official answer: D

Potential difference is energy transferred per unit charge: $V = E/Q$. The positive charge moves through the appliance from the higher to the lower potential, so $E = QV = 2(12) = 24$ J is supplied to it. Answer: D.

Question 27

A solenoid is tightly wound around a smooth horizontal glass tube as shown. A movable aluminium ring is threaded on the right side of the tube. Initially the ring is at rest and the 2-way switch S is in open circuit. In what direction would the ring move at the moment when S is connected in turns to terminal 1 and terminal 2?



- A. Connected to (1): left; connected to (2): left
- B. Connected to (1): right; connected to (2): right
- C. Connected to (1): right; connected to (2): left
- D. Connected to (1): left; connected to (2): right

Official answer: B

When either cell is connected from the open-circuit state, the magnetic flux through the ring starts increasing. Lenz's law gives a ring current whose field opposes this increase, producing repulsion from the solenoid. Reversing the supply reverses both relevant polarities, so the ring moves right in either case. Answer: B.

Question 28

Which of the following statements about domestic circuits is/are correct?

- (1) The live (L) wire is sometimes positive and sometimes negative with respect to the neutral (N) wire.
- (2) The kilowatt-hour meter measures the total power consumed by the domestic electrical appliances.
- (3) With more electrical appliances connected, the total resistance of the domestic circuit becomes smaller.

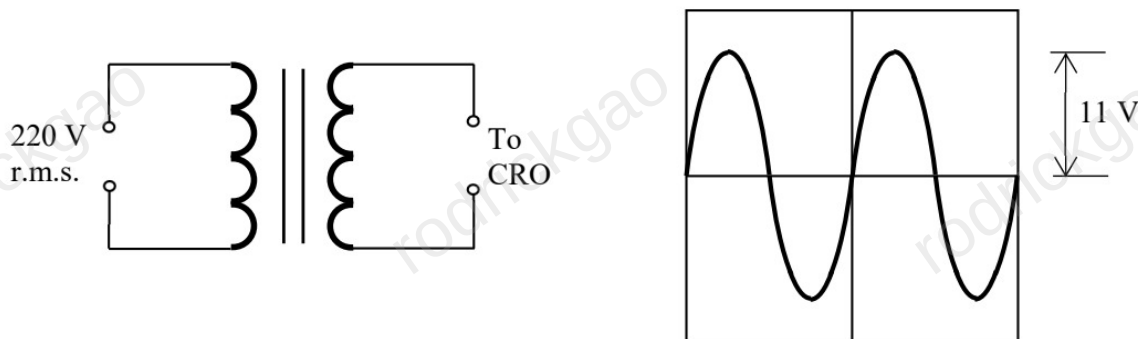
- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: C

The live–neutral voltage is alternating, so its sign reverses. A kilowatt-hour meter records energy, not power. Domestic appliances are connected in parallel; adding another branch increases total conductance and reduces equivalent resistance. Thus (1) and (3) are correct. Answer: C.

Question 29

A transformer is connected to a.c. mains of 220 V r.m.s. value. The peak value of the output voltage displayed on a CRO is 11 V. If the transformer has 2000 turns in its primary coil, find the number of turns in the secondary coil.



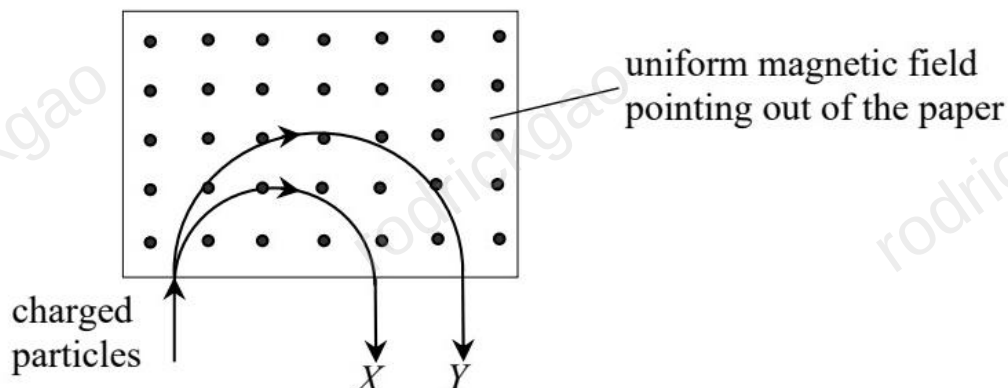
- A. 71
- B. 100
- C. 141
- D. 200

Official answer: A

Use the same voltage convention on both coils. The secondary r.m.s. voltage is $11/\sqrt{2}$ V, whereas the stated primary voltage is already r.m.s. The turns ratio gives $N_s = 2000(11/\sqrt{2})/220 = 70.7 \approx 71$ turns. Answer: A.

Question 30

The figure shows two charged particles X (mass m_X) and Y (mass m_Y) which enter a region of uniform magnetic field pointing out of the paper. X and Y, having the same speed and charge, move along the paths in the plane of the paper as shown below.



X and Y are

- A. Positively charged and $m_X > m_Y$
- B. Positively charged and $m_X < m_Y$
- C. Negatively charged and $m_X > m_Y$
- D. Negatively charged and $m_X < m_Y$

Official answer: B

Initially the particles move upwards and curve right in a field out of the page. The force direction $q(\mathbf{v} \times \mathbf{B})$ therefore identifies positive charges. Since $r = mv/(|q|B)$, equal speeds and charges give $r \propto m$. X has the smaller radius, so $m_X < m_Y$. Answer: B.

Question 31

A radioactive source contains a radioactive nuclide X which decays to become a stable nuclide Y. Besides X there is no other radioactive nuclides in this source. Which of the following statements must be correct after a time of one half-life of X has elapsed?

- (1) The number of nuclides of X decreases to about half of its initial value.
- (2) The number of nuclides of Y increases to about twice of its initial value.
- (3) The activity of the radioactive source falls to about half of its initial value.

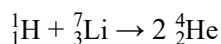
- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

Official answer: C

After one half-life, approximately half the original radioactive X nuclei remain. Since $A = kN_X$ and k is unchanged, the source activity also halves. Y gains the nuclei produced by decay, but its initial number is unspecified, so its number need not double. Thus (1) and (3) are correct. Answer: C.

Question 32

When a proton (${}^1_1\text{H}$) bombards a lithium nucleus (${}^7_3\text{Li}$), two α particles (${}^4_2\text{He}$) are produced.



The energy released in this nuclear reaction is 17.32 MeV. Find the mass of a lithium nucleus.

Given: mass of ${}^1_1\text{H} = 1.0078 \text{ u}$ mass of ${}^4_2\text{He} = 4.0026 \text{ u}$

- A. 6.9788 u
- B. 6.9974 u
- C. 7.0017 u
- D. 7.0160 u

Official answer: D

Energy release means the reactants have the greater total mass. The mass defect is $\Delta m = 17.32/931 = 0.01860 \text{ u}$. Hence $m_{\text{Li}} + 1.0078 - 2(4.0026) = 0.01860$, giving $m_{\text{Li}} = 7.0160 \text{ u}$. The released energy must be added when reconstructing the reactant mass. Answer: D.

Question 33

Which statement about nuclear fission is correct?

- A. It involves two nuclei combined together.
- B. All nuclear fission reactions are spontaneous.
- C. Rate of fission reaction is temperature dependent.
- D. Fission of heavy nuclei yields products that are more stable in terms of energy.

Official answer: D

Fission splits a heavy nucleus into lighter products. In an energy-releasing fission, the products have greater binding energy per nucleon and are more stable. Fission can be induced rather than spontaneous; an ordinary temperature change is not what controls the nuclear reaction in this question. Answer: D.